



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Practical No.: 9

Aim –

Implement localStorage and sessionStorage to save user preferences like theme settings.

Case Study: Create a Schedule Planner for the given table and show an alert box on click event.

Software/Tools Required –

- VSCode
- Chrome
- HTML5
- CSS3
- Javascript (ES6)

Experiment Program Code –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">

  <title>Schedule Selection</title>

  <style>

    /* -----
    GENERAL PAGE STYLE
    ----- */

    body {
      font-family: "Segoe UI", Arial, sans-serif;
      font-size: 17px;
      background-color: #eef4f8;
      color: #263238;
```

```

    margin: 30px;
}

/* -----
   SCHEDULE TABLE
   ----- */

table {
    width: 100%;
    border-collapse: collapse;
    background-color: #ffffff;
    box-shadow: 0 4px 12px rgba(0,0,0,0.08);
}

th {
    background-color: #1f3c88;
    color: white;
    font-size: 18px;
    padding: 14px;
    text-align: left;
}

td {
    padding: 13px;
    border-bottom: 1px solid #d6dee5;
    font-size: 17px;
}

.schedule {
    color: #1f3c88;
    background-color: #ffffff;
    transition: 0.2s;
}

.schedule:hover {
    background-color: #e3f2fd;
    color: #0d47a1;
    cursor: pointer;
}

/* -----
   SAVED DATA
   ----- */

#saved {
    font-size: 18px;
    color: #1565c0;
    background-color: #e8f5e9;
    padding: 15px;
    border-radius: 8px;
    margin-top: 20px;
    line-height: 1.7;
}

/* -----
   CLEAR BUTTON
   ----- */

#clearBtn {
    font-size: 16px;
    color: white;
    background-color: #d32f2f;
    border: none;
    padding: 10px 20px;
    border-radius: 6px;
    cursor: pointer;
}

```

```

        margin-top: 15px;
    }

    #clearBtn:hover {
        background-color: #b71c1c;
    }
}

</style>
</head>

<body>

    <h1>Schedule Selection</h1>

    <table>

        <thead>
            <tr>
                <th>Day</th>
                <th>Schedule</th>
                <th>Topic</th>
            </tr>
        </thead>

        <tbody>

            <tr class="schedule"
                data-day="Monday"
                data-schedule="10:00 AM - 11:00 AM"
                data-topic="Introduction">
                <td>Monday</td>
                <td>10:00 AM - 11:00 AM</td>
                <td>Introduction</td>
            </tr>

            <tr class="schedule"
                data-day="Tuesday"
                data-schedule="11:00 AM - 12:00 PM"
                data-topic="Planning">
                <td>Tuesday</td>
                <td>11:00 AM - 12:00 PM</td>
                <td>Planning</td>
            </tr>

            <tr class="schedule"
                data-day="Wednesday"
                data-schedule="12:00 PM - 01:00 PM"
                data-topic="Implementation">
                <td>Wednesday</td>
                <td>12:00 PM - 01:00 PM</td>
                <td>Implementation</td>
            </tr>

            <tr class="schedule"
                data-day="Thursday"
                data-schedule="02:00 PM - 03:00 PM"
                data-topic="Testing">
                <td>Thursday</td>
                <td>02:00 PM - 03:00 PM</td>
                <td>Testing</td>
            </tr>

            <tr class="schedule"
                data-day="Friday"
                data-schedule="03:00 PM - 04:00 PM"
                data-topic="Review">

```

```

        <td>Friday</td>
        <td>03:00 PM - 04:00 PM</td>
        <td>Review</td>
    </tr>

</tbody>

</table>

<div id="saved">
    No saved preferences.
</div>

<button id="clearBtn">
    Clear Preferences
</button>

<script>

    // -----
    // DOM ACCESS
    // -----

    let schedules =
        document.querySelectorAll(".schedule");

    let saved =
        document.getElementById("saved");

    let clearBtn =
        document.getElementById("clearBtn");

    // -----
    // LOAD SAVED DATA
    // -----

    let savedDay =
        localStorage.getItem("day");

    let savedSchedule =
        localStorage.getItem("schedule");

    let savedTopic =
        sessionStorage.getItem("topic");

    // -----
    // DISPLAY SAVED DATA
    // -----

    if (
        savedDay !== null &&
        savedSchedule !== null
    ) {

        saved.style.color = "green";

        saved.innerHTML =

            "<b>Selected Day:</b> " +
            savedDay +
            "<br>" +

            "<b>Selected Schedule:</b> " +
            savedSchedule +
            "<br>" +

```

```

        "<b>Current Topic:</b> " +
        (savedTopic || "Not selected");

    }

    // -----
    // CLICK EVENT
    // -----

    schedules.forEach(function(row) {

        row.addEventListener("click", function() {

            // Get Day

            let day =
                row.dataset.day;

            // Get Schedule

            let schedule =
                row.dataset.schedule;

            // Get Topic

            let topic =
                row.dataset.topic;

            // -----
            // ALERT BOX
            // -----

            alert(
                "Day: " + day +
                "\nSchedule: " + schedule +
                "\nTopic: " + topic
            );

            // -----
            // CHANGE TOPIC
            // -----

            let newTopic =
                prompt(
                    "Enter new topic:",
                    topic
                );

            // Check if user entered a topic

            if (
                newTopic !== null &&
                newTopic.trim() !== ""
            ) {

                newTopic =
                    newTopic.trim();

                // Update topic in HTML

                row.dataset.topic =
                    newTopic;

                // Update topic displayed in table

                row.cells[

```

```

        row.cells.length - 1
    ].innerHTML =
        newTopic;

    // Update topic variable

    topic =
        newTopic;

}

// -----
// LOCAL STORAGE
// -----

localStorage.setItem(
    "day",
    day
);

localStorage.setItem(
    "schedule",
    schedule
);

// -----
// SESSION STORAGE
// -----

sessionStorage.setItem(
    "topic",
    topic
);

// -----
// DISPLAY SELECTED DATA
// -----

saved.style.color =
    "green";

saved.innerHTML =

    "<b>Selected Day:</b> " +
    day +
    "<br>" +

    "<b>Selected Schedule:</b> " +
    schedule +
    "<br>" +

    "<b>Current Topic:</b> " +
    topic;

// -----
// HIGHLIGHT SELECTED ROW
// -----

schedules.forEach(
    function(item) {

        item.style.backgroundColor =
            "";

    }
);

```

```

        row.style.backgroundColor =
            "#dbeafe";

    });

});

// -----
// CLEAR PREFERENCES
// -----

clearBtn.addEventListener(
    "click",
    function() {

        // Remove Day

        localStorage.removeItem(
            "day"
        );

        // Remove Schedule

        localStorage.removeItem(
            "schedule"
        );

        // Remove Topic

        sessionStorage.removeItem(
            "topic"
        );

        // Display message

        saved.style.color =
            "red";

        saved.innerHTML =
            "Saved preferences cleared.";

        // Remove highlighting

        schedules.forEach(
            function(row) {

                row.style.backgroundColor =
                    "";

            }
        );

    }
);

</script>

</body>
</html>

```

Output –

Experiment No. 9

localStorage & sessionStorage – User Preferences SEJAL LENDE 24070521035

Theme Preference localStorage

Select Theme:

☒  Light Mode ☐  Dark Mode

Theme is saved in **localStorage** and persists even after closing the browser.

Topic Selection sessionStorage

Select a Topic:

HTML

Topic is saved in **sessionStorage** and is cleared when you close this tab.

Storage Status & Actions

Show Stored Values

Clear Preferences

localStorage → **theme:** light
sessionStorage → **topic:** HTML

Experiment No. 9

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Select Theme:

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Topic Selection sessionStorage

Select a Topic:

HTML

Topic is saved in **sessionStorage** and is cleared when you close this tab.

Storage Status & Actions

Show Stored Values

Clear Preferences

localStorage → **theme:** dark

sessionStorage → **topic:** HTML

Case Study Title –

Create a Schedule Planner for the given table and show an alert box on click event.

Case Study Program Code –

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Schedule Planner – Seminar Timetable</title>
  <style>
    * { box-sizing: border-box; margin: 0; padding: 0; }

    body {
      font-family: 'Segoe UI', system-ui, sans-serif;
      background: #f0f4f8;
      min-height: 100vh;
      padding: 30px 16px;
      color: #1a1a2e;
    }

    h1 {
      text-align: center;
      font-size: 1.6rem;
      margin-bottom: 6px;
    }

    .subtitle {
      text-align: center;
      color: #666;
      font-size: 0.9rem;
      margin-bottom: 28px;
    }

    .table-wrapper {
      max-width: 720px;
      margin: 0 auto;
      overflow-x: auto;
      border-radius: 12px;
      box-shadow: 0 4px 16px rgba(0,0,0,0.08);
      background: #fff;
    }

    table {
      width: 100%;
      border-collapse: collapse;
```

```

    font-size: 0.95rem;
}

th, td {
    border: 1px solid #ccc;
    padding: 12px 14px;
    text-align: center;
}

thead th {
    background: #2c3e50;
    color: #fff;
    font-weight: 600;
}

thead th.seminar {
    background: #1a5276;
    font-size: 1.05rem;
    letter-spacing: 0.5px;
}

/* Day column */
td.day {
    font-weight: 600;
    background: #eaf2f8;
    vertical-align: middle;
}

/* Time cells – matching the colours from the image */
.begin-yellow { background: #f9e79f; }
.end-purple { background: #d2b4de; }
.begin-yellow2 { background: #f9e79f; }
.end-yellow { background: #f9e79f; }
.begin-green { background: #abebc6; }
.end-green { background: #abebc6; }
.end-purple2 { background: #d2b4de; }

/* Topic cells */
td.topic {
    text-align: left;
    padding-left: 16px;
}

/* Clickable rows */
tbody tr {
    cursor: pointer;
    transition: background 0.2s, transform 0.1s;
}
tbody tr:hover {
    filter: brightness(0.97);
}

```

```

        outline: 2px solid #3498db;
        outline-offset: -2px;
    }
    tbody tr:active {
        transform: scale(0.995);
    }

    .hint {
        text-align: center;
        margin-top: 18px;
        font-size: 0.85rem;
        color: #888;
    }

    footer {
        text-align: center;
        margin-top: 32px;
        font-size: 0.8rem;
        color: #aaa;
    }
</style>
</head>
<body>

<h1>Seminar Schedule Planner</h1>
<p class="subtitle">Click any row to view full details in an Alert Box</p>

<div class="table-wrapper">
    <table>
        <thead>
            <tr>
                <th rowspan="2">Day</th>
                <th colspan="2" class="seminar">Seminar</th>
            </tr>
            <tr>
                <th>Schedule</th>
                <th>Topic</th>
            </tr>
            <tr>
                <th></th>
                <th>Begin &nbsp;  |&nbsp;  End</th>
                <th></th>
            </tr>
        </thead>
        <tbody>
            <!-- Monday -->
            <tr onclick="showAlert('Monday', '8:00 a.m.', '5:00 p.m.', 'Introduction to
XML\nValidity: DTD and Relax NG')">
                <td class="day" rowspan="1">Monday</td>
                <td>

```



```

        <!-- Wednesday -->
        <tr onclick="showAlert('Wednesday', '8:00 a.m.', '12:00 p.m.', 'XSL Formatting
Objects')">
            <td class="day">Wednesday</td>
            <td>
                <span class="begin-yellow" style="padding:4px 8px;border-radius:4px;">8:00
a.m.</span>
                &nbsp;–&nbsp;
                <span class="end-green" style="padding:4px 8px;border-radius:4px;">12:00
p.m.</span>
            </td>
            <td class="topic">XSL Formatting Objects</td>
        </tr>
    </tbody>
</table>
</div>

<p class="hint">💡 Click on any row to see the schedule details in an alert box.</p>

<footer>Schedule Planner &mdash; Click Event with alert()</footer>

<script>
/**
 * Shows an alert box with the schedule details
 * Called on click of each table row
 */
function showAlert(day, begin, end, topic) {
    const message =
        "📅 Seminar Schedule Details\n" +
        "_____\n" +
        "Day   : " + day + "\n" +
        "Begin : " + begin + "\n" +
        "End   : " + end + "\n" +
        "Topic : " + topic;

    alert(message);
}
</script>
</body>
</html>

```

Output –

127.0.0.1:5500 says

 Seminar Schedule Details

Day : Tuesday
Begin : 2:00 p.m.
End : 5:00 p.m.
Topic : XSL Transformations

OK

Day		
Monday	8:00 a.m. – 5:00 p.m.	Introduction to XML Validity: DTD and Relax NG
Tuesday	8:00 a.m. – 11:00 a.m.	XPath
	11:00 a.m. – 2:00 p.m.	—
	2:00 p.m. – 5:00 p.m.	XSL Transformations
Wednesday	8:00 a.m. – 12:00 p.m.	XSL Formatting Objects

 Click on any row to see the schedule details in an alert box.

Schedule Planner — Click Event with alert()

127.0.0.1:5500 says

 Seminar Schedule Details

Day : Monday
Begin : 8:00 a.m.
End : 5:00 p.m.
Topic : Introduction to XML
Validity: DTD and Relax NG

OK

Day		
Monday	8:00 a.m. – 5:00 p.m.	Validity: DTD and Relax NG
Tuesday	8:00 a.m. – 11:00 a.m.	XPath
	11:00 a.m. – 2:00 p.m.	—
	2:00 p.m. – 5:00 p.m.	XSL Transformations
Wednesday	8:00 a.m. – 12:00 p.m.	XSL Formatting Objects

 Click on any row to see the schedule details in an alert box.

Schedule Planner — Click Event with alert()

Conclusion –

The experiment successfully demonstrated **web storage and JavaScript click events** through theme preferences and a Schedule Planner.